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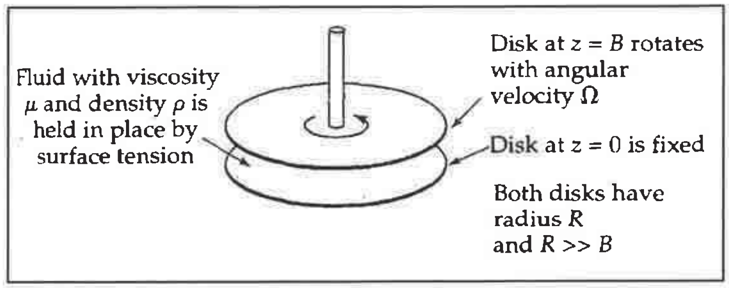
TRANSPORT PHENOMENA I, CHE 311

MIDTERM EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

PARALLEL-DISK VISCOMETER

A fluid is placed in the gap between two horizontal circular disks of radius R. Disks are separated by a small distance B. The lower disk is held fixed. The upper disk is rotating at a constant angular velocity Ω.



Postulates:

- Newtonian fluid with constant density ρ and constant viscosity μ.
- $v_r = 0$ and $v_z = 0$.
- Pressure in the fluid is not a function of θ.
- Creeping flow assumption: The flow is very slow and the acceleration term $\frac{D[v]}{Dt}$ in the equation of motion is negligible.

10 1. Using the continuity equation, prove v_θ is not a function of θ at steady-state conditions.

20 2. Using the proper equation of motion at steady-state conditions, write down and simplify a partial differential equation for v_θ as a function of r and z.

5 3. Assuming v_θ may be written as r multiplied by a function of z,

$$v_\theta(r, z) = r\, f(z)$$

, convert the partial differential equation for v_θ to an ordinary differential equation for f.

0 4. Knowing $v_\theta(r, z = 0) = 0$ and $v_\theta(r, z = B) = r\,\Omega$, write down two boundary conditions for f, solve the ordinary differential equation derived in part 1.3. and evaluate the integration constants.

0 5. One can estimate the viscosity of the fluid by measuring the torque T_z required to rotate the upper disk at a constant angular velocity. Using the velocity profile derived in part 1.4., find an equation for the viscosity in terms of T_z, Ω, B and R. Hint: torque exerted by the liquid on unit area of the upper disk is the product of the radius r (lever arm) and viscous stress in θ-direction.

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1. $\frac{\partial \rho}{\partial t} + (\nabla \cdot \rho V) = 0$

$$\frac{\partial \rho}{\partial t} + \underbrace{\rho\left(\frac{\partial v_\theta}{\partial r} + \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_\theta}{\partial z}\right)}= 0$$

x

due to constant density, this whole term goes to 0, and $\frac{\partial v_\theta}{\partial \partial}$ doesn't exist.

2. v_θ as a function of r and z at S.S.

$\frac{\partial V}{\partial T} = 0$

$$\cancel{\rho\left(\frac{\partial v_\theta}{\partial t} + V_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta} + V_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r}\right)} = -\cancel{\frac{1}{r}\frac{\partial p}{\partial \theta}} + \mu\Big[$$

$$\underbrace{\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_\theta)\right)} + \cancel{\frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2}} + \frac{\partial^2 v_\theta}{\partial z^2} + \cancel{\frac{2}{r^2}\frac{\partial v_r}{\partial \theta}}\Big] + \rho g_\theta$$

✓

$$0 = \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_\theta)\right) + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$ ✓

$$0 = \frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{\partial^2 v_\theta}{\partial z^2}$$

$$0 = v_\theta \frac{\partial}{\partial r}\left(\frac{1}{r}\right) + \frac{\partial^2 v_\theta}{\partial z^2}$$

$$0 = v_\theta \ln(r) + \frac{\partial^2 v_\theta}{\partial z^2}$$

$$\boxed{-\frac{\partial^2 v_\theta}{\partial z^2} = v_\theta \ln(r)}$$

3. $-\frac{\partial^2 f}{\partial z^2} = r(fr)\ln(r)$

$$\boxed{-\frac{d^2 f}{dz^2}\cdot\frac{1}{B^2} = f r^2 \ln(r)}$$

4. $v_\theta(r, 0) = 0$ $\quad v_\theta(r, B) = r\Omega$

$f(1, z) = 0$ $\quad f(1, B) = \Omega$

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